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IMT 574 A: Introduction to Machine Learning
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1. **Why would you advocate for bringing transparency or explainability in an ML/AI system? What are the advantages to the users as well as the organization providing that system?**

Explainability is essential because, as Arrieta et al. note, black-box ML models pose risks when their decisions cannot be easily justified (83). For users, transparency increases confidence by clarifying why a model made a particular decision, helping them detect potential bias and better understand outcomes that affect them. For organizations, explainability supports regulatory compliance, strengthens trust, and helps identify issues such as dataset bias or model vulnerabilities. Overall, transparency promotes responsible use of AI by improving understanding, fairness, and accountability.

2. **What is a "fair" system? Take an example of an ML/AI system and tell us what achieving "fairness" would look like.**

A “fair” system is one in which model decisions do not systematically disadvantage individuals or groups, and where the reasoning behind those decisions is transparent enough to allow ethical evaluation. As Arrieta et al. explain, explainability plays a key role in assessing fairness because it makes the relationships driving a model’s output visible for scrutiny (87).

An example scenario is an ML system used to assess a consumer’s creditworthiness. Achieving fairness would mean ensuring the model does not treat applicants differently based on factors like race, gender, or disability, and that its decision-making process can be clearly understood. This might involve using interpretable features, providing visual explanations such as decision paths or feature importance, and demonstrating that applicants with similar financial profiles receive similar outcomes.

3. **How do you see concepts like bias, fairness, accountability, and transparency-related?**

Bias, fairness, accountability, and transparency are deeply interconnected. Transparency enables us to see how a model reaches its decisions, which makes it possible to identify bias and evaluate whether the system is behaving fairly. Fairness depends on being able to detect and mitigate biased patterns in data or model behavior, something that is only achievable when explanations are available.

While fairness and transparency are things we aim to build into the model itself, accountability involves the responsibilities of designers, developers, and users. As Arrieta et al. emphasize, human oversight is essential for preventing biases from being embedded in models and for ensuring systems are aligned with ethical principles (103). Accountability ensures that organizations and practitioners use transparency and fairness appropriately and intervene when model behavior is harmful or discriminatory.

4. **Give an example of a system that is biased but fair.**

An e-commerce recommendation system, such as Amazon's, can be an example of a system that is biased but fair. It is biased because it uses a user's past browsing and purchase patterns to make assumptions about what products they may prefer, which creates a statistical bias toward certain items. However, it can still be fair if these recommendations are based only on non-sensitive behavioral data and produce similar quality suggestions for all users regardless of attributes like gender or race. In this case, so long as the system applies the same criteria consistently across users, bias supports personalization rather than discrimination.

Resources

Barredo Arrieta, A., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., García, S., Gil-Lopez, S., Molina, D., Benjamins, R., Chatila, R., & Herrera, F. (2020). Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion, 58*, 82-115. https://doi.org/10.1016/j.inffus.2019.12.012